

Multi-Scale Target Detection in SAR Image Based on Visual Attention Model

Zhaocheng Wang,[#] Lan Du, Fei Wang, Hongtao Su and Yu Zhou

National Laboratory of Radar Signal Processing

Xidian University

Xi'an, 710071, China

dulan@mail.xidian.edu.cn

Abstract—This paper proposes a novel method for synthetic aperture radar (SAR) target detection by using multi-scale SAR images based on visual attention model, which can automatically find the vehicle targets from the complicated background with clutters such as trees and buildings. In our method, firstly, a saliency map is obtained from a Gaussian pyramid of the original SAR image, where the image scales are selected based on the prior size information of the targets to be detected in the image. Secondly, we use the method based on shifts of the focus of attention (FOA) in the saliency map to get a binary image. Finally, the clustering algorithm based on the prior length of targets is employed to extract the target candidate chips in the binary image. In the experiment based on the real SAR image, we compare the proposed method with the classical constant false alarm rate (CFAR) target detection method, which indicates that our method can detect vehicle targets in the image more quickly and with fewer false alarms.

Index Terms—Target detection, synthetic aperture radar (SAR) image, visual attention, clustering, constant false alarm rate (CFAR), saliency map.

I. INTRODUCTION

Lincoln Laboratory has developed a system of synthetic aperture radar (SAR) automatic target recognition (ATR) and target detection is a first and important step for this system [1]. Constant false alarm rate (CFAR) is a conventional method of target detection in SAR images [2], [3], but it has some disadvantages. Firstly, the proper clutter statistical model is difficult to choose and an inappropriate model will result in false alarms. Secondly, the speed of CFAR is too slow for the application of the real-time SAR ATR system.

It is noted that the ability of human visual attention system to detect targets from the background in optical images is extraordinarily fast and reliable [4], [5], [6], [10]. In 1998, Itti et al. [4] proposed a model of saliency-based visual attention and then it becomes the basis of later models [7], however, the Itti model is purely data-driven and it does not consider the detection task. Recently, the idea of visual attention which using single scale image to generate saliency map has been introduced into SAR images and obtained some good performance [8],[9], of which Hou et al. [9] introduced some top-down factors to water segmentation. However, these methods are effective in the simple scenes, such as ship detection in the water regions. When our task is to detect

targets in the complicated scenes, such as vehicle targets in the urban regions with clutters such as trees, buildings and other targets, the classical visual attention model loses effectiveness.

To find the vehicle targets from the complicated background and overcome the defects of CFAR, in this paper, we propose a novel method for target detection by using multi-scale SAR images based on visual attention model, which make full use of target prior knowledge about the detection task. To pop out the targets while obscure the background and clutters, task-dependent scales are chosen from the Gaussian pyramid of the original SAR image based on the prior knowledge of targets. The salient pixels are obtained with the operation called “shifts of the focus of attention (FOA)” in the saliency map, and then the clustering algorithm based on the prior length of targets is applied to merge the salient regions. Finally the number of pixels of every cluster is used to discriminate the targets from clutters to eliminate the false alarms. In the experiment based on the real SAR image, we compare the proposed method with the classical CFAR. The results indicate that our method can detect vehicle targets in the image more quickly and with fewer false alarms.

The remainder of this paper is structured as follows: Section II briefly introduces the classic visual attention model. Section III describes the novel SAR target detection method. Section IV compares the experimental results of the proposed SAR target detection method with CFAR. Conclusion is given in the last section.

II. CLASSICAL VISUAL ATTENTION MODEL

The main process of the classic visual attention model (Itti model) for optical image is as follows [4], [10]. Firstly, an input image is decomposed into three channels: intensity, color and orientation. Secondly, an operation called “center-surround” produces a sequence of feature maps. These feature maps are then normalized and combined across scales to create conspicuity maps for each channel. The normalized conspicuity maps are further linearly combined to form the saliency map. Then, the FOA areas are located based on the saliency map. Finally, the operation of winner-take-all is employed to find the most significant area from the saliency map, and the operation of inhibition-of-return is used to guarantee that the area would not be focused any more.

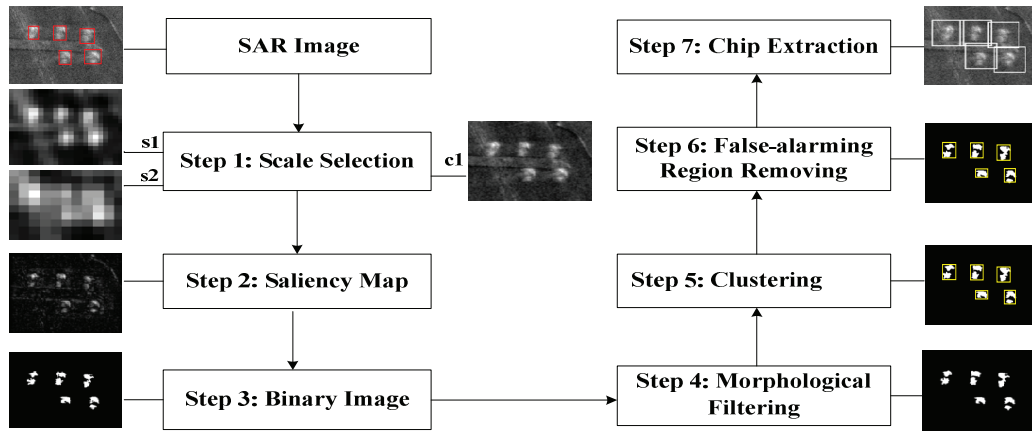


Fig. 1 Flowchart of the proposed SAR image target detection method

III. PROPOSED SAR DETECTION METHOD

The classical visual attention model is proposed for the optical images. The imaging mechanism of optical and SAR images are different from each other. For SAR images, we consider only using the intensity channel to detect targets. In this section, we first give an overview of the proposed SAR target detection method, which mainly including seven processing steps. Then, we describe the crucial steps of the proposed algorithm in detail.

A. Overview of the Proposed Algorithm

The flowchart of the proposed SAR image target detection method is given in Fig.1. An example on a small SAR image is shown along each step for the better illustration purpose. The scene of the SAR image used in this example is rather simple, and more complicated scene will be employed in the experiment section. In the first image, the ground truth of the target regions in this example is marked with red boxes. Explanation for each step is given in the following.

Step 1: Itti et al. use an operation called “center-surround” [4], which calculates the difference between the fixed center and surround based on the Gaussian pyramid of the original image to construct the saliency map. To pop-out the targets of interest while obscure the background and clutters (such as trees, buildings and other targets), in this paper, we select the task-dependent scales from the Gaussian pyramid based on the prior size information of the targets to be detected in Step 1 of our method, rather than using the fixed scales for the operation of center-surround as Itti does. Here we show our selected images from the Gaussian pyramid of our example.

Step 2: Based on the result of the scale selection in step 1, we linearly combine all the images obtained via the operation of center-surround to construct a saliency map as Itti does [4].

Step 3: Since the intensity of the target regions will be larger than its surrounding regions via the appropriate scale selection, the target regions have priority to be detected. In Step 3, we get the binary image marking all the salient pixels

via the operation called “shifts of FOA” [4] on the saliency map.

Step 4: The closing operation of morphological filter is applied to the binary image to connect narrow gaps and fill recesses.

Step 5: In the uniform optical image, the classical visual attention model regards every isolated salient regions as one complete target. Due to the property of SAR image, the salient pixels in the binary image are not capable of forming a complete target region and may be separated into several parts. To join the several parts into a complete target region, it is necessary to employ the clustering algorithm in the binary image. In Step 5, we adopt the clustering method proposed by Gao et al. [3], which groups every two connected regions with their maximum pixel distance within a threshold. Here the prior information about the lengths of the targets to be detected is used as the threshold. The salient pixels are clustered into five categories in this example and they are marked with yellow boxes in this image.

Step 6: To eliminate the false alarms, the clustered regions with the number of pixels, S , smaller than the prior information about the minimum number of pixels of the targets to be detected, S_{min} , or larger than that corresponding to the maximum number of pixels, S_{max} , i.e., $S < S_{min}$ or $S > S_{max}$, are not regarded as the target clusters in the next step. In this example, the number of pixels in each cluster in Step 5 is just within the threshold range, and all of them are regarded as the suspected targets.

Step 7: The target candidate chips are extracted in the original SAR image based on the centroids of target clusters in the binary image. The target detection results in the example are marked with white boxes in this image.

In the following we will explain Steps 1, 3, 5 and 6 in detail.

B. Scale Selection

Figure 2 shows the dyadic Gaussian pyramid of a SAR image example. The nine images in different spatial scales are

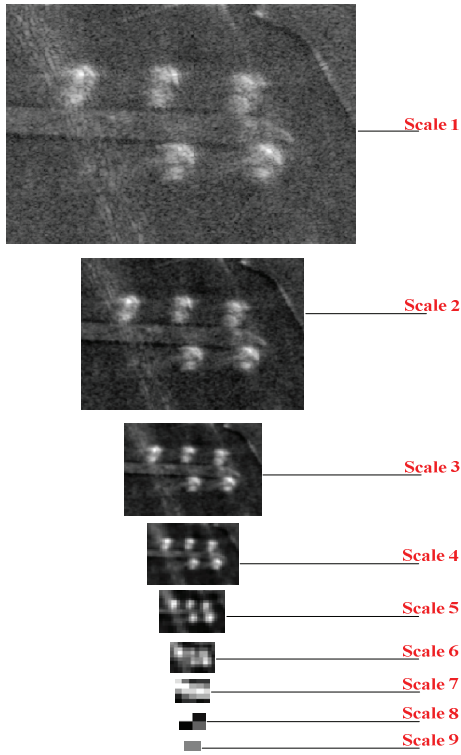


Fig. 2 Gaussian pyramid of SAR image

created by progressively low-pass filtering and subsampling the input SAR image with the reduction factors 1:1 (scale 1, original scale), 1:2 (scale 2), 1:4 (scale 3), 1:256 (scale 9) for both of horizontal and vertical dimensions. In this example, we get $n=9$ multi-scale images and the images are expressed as $I_\sigma = \{I_1, I_2, I_3, \dots, I_9\}$. In [4], Itti et al. used an operation called center-surround which is implemented as the difference between the center pixel at scales $c \in \{2, 3, 4\}$ in the Gaussian pyramid and the corresponding surround pixel at scales $s = c + \delta$ with $\delta \in \{3, 4\}$, i.e., at scales $I_2 - I_5$, $I_2 - I_6$, $I_3 - I_6$, $I_3 - I_7$, $I_4 - I_7$, $I_4 - I_8$. Note that all the images I_σ in the Gaussian pyramid are interpolated to scale 1 before the operation of center-surround. As shown in [4], Itti used the fixed center ($c \in \{2, 3, 4\}$) and surround scales ($s = c + \delta$ with $\delta \in \{3, 4\}$) for different images with different detection tasks. However, different scales for center-surround will highlight different size of objects in the image and lead different priority to be detected, therefore, the fixed center and surround scales are inappropriate for different images with different detection tasks. We find that at some surround scales, the large area clutters, e.g., trees and buildings, are clear, while the small vehicle targets are fused into the background in the SAR image. So the appropriate scale selection becomes an important step to detect the targets of interest from the complicated background and clutters. We explain how to select the center and surround scales for center-surround in the following.

Firstly, the center scales c should guarantee the targets in the SAR image are clearly enough to be detected. To achieve this goal, we can choose one or two fine scales, i.e., $c = \{1, 2\}$.

Secondly, the surround scales s should ensure the target coarse enough in the SAR image while the large area of the background and clutters are still unambiguous. To achieve this goal, the surround scales s are selected based on the prior size information of the target to be detected. Assume that the real length and width of the target of interest are L and W , respectively. Both the range and azimuth resolution of the SAR image are ΔR . Then the length l and the width w of the target in the SAR image are

$$l = L / \Delta R, \quad (1)$$

$$w = W / \Delta R. \quad (2)$$

If we define

$$M = \min(l, w), \quad (3)$$

then the best surround scale s_{best} should satisfy

$$M / 2^{(s_{best}-1)} \approx 1. \quad (4)$$

In order to make the algorithm more robust, one or two surround scales close to s_{best} can be selected.

C. Generation of Binary Image

The target pixels are highlighted and most of their intensities are higher than those of the background and clutters in the saliency map I_s . The maximum value of the saliency map indicates the most salient location, to which FOA should be directed. At a FOA location, the region-growing segmentation algorithm [11] is employed to generate a salient region. To guarantee the salient area after segmentation would not be focused again, the pixels of the current salient region are set to be zero in the saliency map I_s for the next shift of FOA operation. Then the maximum value in the saliency map I_s will become smaller and smaller with shifts of FOA. Here the termination condition for shifts of FOA is

$$A_m < T, A_m = \max(I_s). \quad (6)$$

In Eq.(6), the threshold T is calculated as

$$T = \alpha + K \times \beta, \quad (7)$$

where α and β are the mean and standard deviation of the saliency map, respectively, and K is the constant coefficient that can control the false alarm. It is worth emphasizing that the larger K is, the fewer false alarms it will generate, and vice versa. In this paper K is set to be 7.3, which guarantees high detection rate and low false alarm rate.

The operation "shifts of FOA" is implemented in the order of decreasing saliency in the saliency map until the maximum value of the saliency map A_m satisfies Eq.(6). After shifts of FOA, a binary image I_b marking all the salient regions is generated.

D. Clustering and False Alarm Elimination

To join the several parts into a complete target region in the binary image I_b , we employ the clustering algorithm proposed by Gao et al. [3], which clusters every two



Fig. 3 Original SAR image, with target regions (vehicles) marked by yellow rectangles.

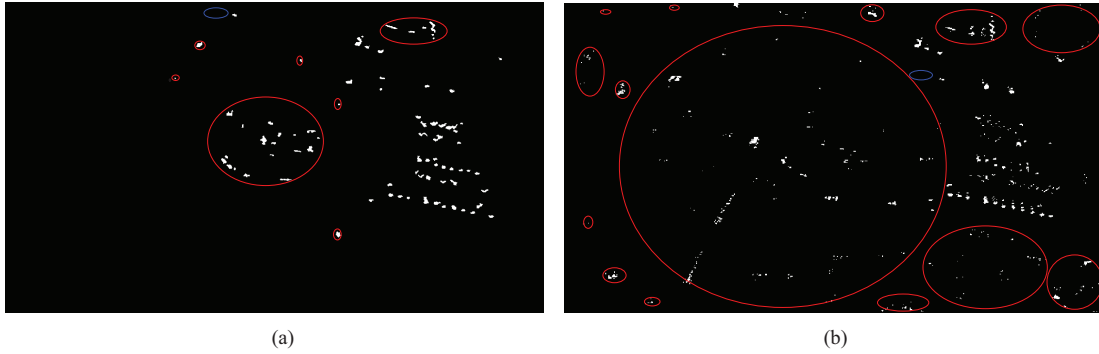


Fig. 4 Binary image. (a) Binary image after false-alarming region removing acquired by our method. (b) Binary image after false-alarming region removing acquired by CFAR.

connected regions with their maximum pixel distance within a threshold $d_{\max}$. According to Eq.(1) and Eq.(2), the threshold $d_{\max}$ is calculated as follows:

$$d_{\max} = \sqrt{l^2 + w^2}. \quad (8)$$

In practice, to eliminate the false alarms, the maximum number of pixels of the targets to be detected, $S_{\max}$, is represented as:

$$S_{\max} = L \times W / (\Delta R \times \Delta R), \quad (9)$$

where L and W are the real length and width of the target of interest respectively, ΔR is the range and azimuth resolution of the SAR image. The minimum number of pixels of the targets to be detected $S_{\min}$, is determined by $S_{\max}$ and it is calculated as:

$$S_{\min} = S_{\max} / P, \quad P > 1. \quad (10)$$

In this paper, P is set to be 15.

E. Comparison with Classical Visual Attention Model

Our target detection method has two different aspects from the classical visual model. One is that we select the task-dependent scales from the Gaussian pyramid based on the prior size information of the targets to be detected in the SAR image, rather than using the fixed scales for the operation of

center-surround as Itti does [4],[5],[10]. The other one is that we adopt the clustering method proposed by Gao et al. [3] to join the several isolated FOA areas into a complete target region, instead of regarding every isolated FOA area as one complete target in Itti model [4], [5], [10].

IV. EXPERIMENTAL RESULTS

To validate the effectiveness of the proposed SAR image target detection algorithm, a real high-resolution SAR image, with the resolution is $0.1m \times 0.1m$ (the vertical represents the range dimension and the horizontal is the azimuth dimension) and the size is 1638×2510 , is applied into the experiment. The real SAR image is from miniSAR data set, acquired by Sandia National Laboratories of America¹. In the SAR image, the vehicles are the targets we are interested in, and the others, like buildings, trees, and lawns are treated as the clutters. For comparison, we also give the detection results by the conventional target detection method two-parameter CFAR [1], [3], where the false alarm rate is set to 0.47%. For our method, the center scales are $c = \{1, 2\}$, the surround scales are $s = \{4, 5\}$, and the chip size is set as 100×100 .

The experiments are conducted using a PC with Intel(R) Core i5-4590 CPU of 3.3 GHz and memory of 8 GB. The

¹ <http://www.sandia.gov/radar/imagery>

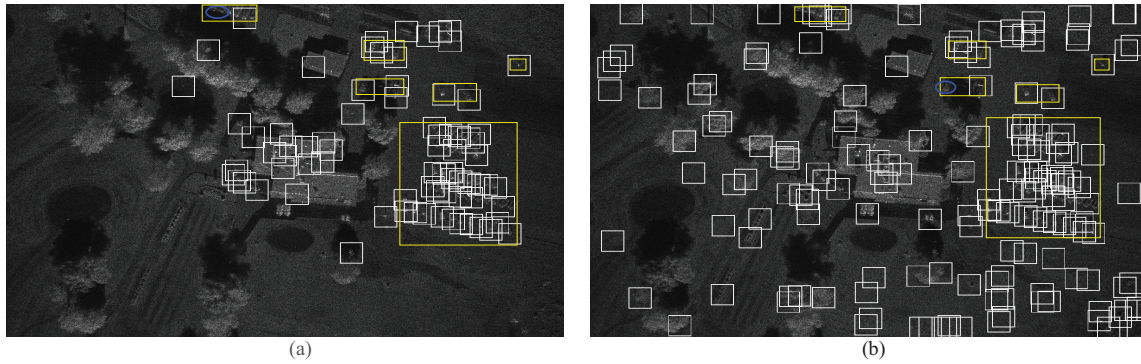


Fig.5 Detection result. (a) Chips extracted by our method. (b) Chips extracted by CFAR.

TABLE I. COMPARISON OF DETECTION RESULTS OF TWO METHODS

	<i>Number of all targets</i>	<i>Number of detected targets</i>	<i>Number of false alarms</i>	<i>Time Cost (seconds)</i>
CFAR	53	52	113	12.6
The proposed method	53	52	25	3.2

program codes are written in MATLAB.

Figure 3 is the original SAR image. In Fig.3 the vehicle target regions are manually marked by the yellow rectangles, which will be used as the ground truth for the following experiment.

Figure 4(a)-(b) are binary images after region removing acquired by our method and by CFAR respectively, where the regions marked with red ellipses are false alarms and the blue ellipses represent the missed alarms. In Fig. 4(a), we can see that the white regions are mainly from the vehicles and buildings, while those in Fig. 4(b) are scattered in the whole image uniformly.

Figure 5(a) is the detection result of our method, and 77 chips denoted by white rectangles are extracted in 3.2 seconds, with one missed alarm marked by the blue ellipse. Fig. 5(b) is the detection result of CFAR, and 165 chips denoted by white rectangles are extracted in 12.6 seconds, with one missed alarm marked by the blue ellipse. The ground truth of the target regions are marked by the yellow rectangles manually. Table 1 gives the quantitative comparison results. It is easy to see from Table 1 that our method is not only much faster than CFAR but also with much fewer false alarms.

V. CONCLUSIONS

A multi-scale target detection method in SAR image based on visual attention model has been presented in this paper. The experimental results show that our SAR target detection method works well, with the help of using the prior size information of the target to be detected by scale selection and clustering. Compared with the conventional target detection method CFAR, the proposed SAR target detection method has higher efficiency and yields fewer false alarms. Nevertheless, as shown in our experiment results, there are still some clutter

chips extracted as target candidate chips via our detection method. In our future work, we will further consider how to discriminate the target chips from the candidate chips in the discrimination step in the SAR ATR system [1].

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